REQUIREMENTS ANALYSIS · TEST STRATEGY · TEST PLAN · PRIORITISED TEST CASES · RELEASE METRICS

GymFlow Win-back Offer: test strategy, test plan and test cases for a churn-reduction feature

How I take a new requirement from analysis to a release decision: questions for the team first, then a risk-based strategy, test cases covering web, mobile and API, and three metrics to judge the release.

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> GymFlow and its requirement are a fictional scenario I wrote for this sample. No real company, product or data is involved.

The requirement in brief

GymFlow is a SaaS platform for gyms and sports clubs: a **web app for staff** (admin and front desk) and a **mobile app for members**. 38% of members cancel within their first 90 days. The **Win-back Offer** detects inactive members close to renewal and sends them a discount through an external mailer API.

Eligibility

- Feature flag `winback_offer_enabled` is on.
- Monthly or Quarterly plan, expiring in 10 days or fewer.
- Fewer than 4 check-ins in the last 28 days.
- Solo plan: 15%. Duo plan: 25% if both members are inactive. Family plan (up to 5, auto-renew): see rules.
- Mailer must answer in under 2 s; errors go to `/api/v2/audit/errors`.

Business rules

B1	Members with an outstanding balance do not receive an offer.
B2	Corporate memberships are excluded.
B3	Maximum one offer every 90 days per member.
B4	Family plans are evaluated on the group's average check-ins.
B5	Auto-renew plans do not receive a discount.
B6	Members who joined less than 45 days ago are excluded.
B7	Offers do not stack with an active promotion.

1. Requirement analysis

Assumptions

- The daily eligibility job runs once at 03:00 in the gym's local time zone; the 28-day window is calendar days ending yesterday.
- A check-in is a QR scan at the turnstile or in the app. A class booking without a check-in does not count.
- "Expires in 10 days or fewer" includes day 10 and excludes memberships that already expired.
- The offer is valid until the membership expiry date. Accepting it after that date does not apply the discount.
- The external mailer is replaced by a mock in the test environment, so failures and slow responses can be simulated.

Risks, ambiguities and questions for the team

ID	Finding	Risk if not resolved	Ask
Q-01	B4 and B5 contradict each other: Family plans renew automatically (no discount) but B4 defines how to evaluate them.	Family members get a discount the business did not intend, or never get one it did.	Product Owner
Q-02	The email copy is fixed at "15% off", but Duo plans get 25%.	Members see one price in the email and another at checkout; support tickets and loss of trust.	Product Owner, Marketing
Q-03	"Low activity" is described both as an average and as "fewer than 4 check-ins in 28 days". Time zone and the exact window are not defined.	Members at the boundary are classified differently by each component.	Product Owner, Backend
Q-04	No behaviour defined when the mailer API fails or takes longer than 2 s: retry, queue, or still create the internal offer?	Offers without an email, or duplicate emails after retries.	Backend, Tech Lead
Q-05	No idempotency rule if the daily job runs twice.	Same member receives two offers, breaking B3.	Backend
Q-06	What happens to pending offers when the flag is switched off?	Members click a valid-looking link that no longer applies a discount.	Product Owner
Q-07	Can the front desk apply a member's pending offer when renewing on their behalf?	In-person renewals lose the discount; inconsistent experience across channels.	Product Owner, Operations
Q-08	Marketing consent is not mentioned: can members who opted out of promotional email receive the offer?	Privacy and compliance exposure.	Product Owner, Legal
Q-09	B1 does not say whether a member who pays the balance becomes eligible again inside the same window.	Lost renewals from members who just settled their debt.	Product Owner

2. Test strategy

Approach

Risk-based. The highest risks are financial and trust-related: a discount given to the wrong member, the wrong percentage, or a duplicate offer. Those areas get the deepest coverage and run first.

Shift-left. The questions above go to refinement before development starts, so the answers become acceptance criteria instead of bugs.

Test levels and types

- **API and integration:** eligibility job, internal offers API, mailer contract, logs.
- **Functional end to end:** email, deep link, renewal in the mobile app, staff web app views.
- **Business rules:** decision table and boundary values.
- **Negative and resilience:** mailer errors, timeouts, double runs.
- **Security:** permissions by role.
- **Performance:** mailer response time under a realistic batch.
- **Regression:** regular renewal, pricing and promotions.

Scope

In: eligibility, offer creation, mailer integration, email content, acceptance and renewal, both apps, roles, logs.

Out: the mailer provider's own delivery rates and the marketing design of the email, owned by Marketing.

Environment, data and tools

Staging with the mailer replaced by a mock that can return 201, 4xx, 5xx and slow responses. Seeded members at each boundary (check-ins 3/4/5, expiry 9/10/11 days, joined 44/45 days, last offer 89/90 days). Postman for API checks, SQL to verify offers and logs, Chrome DevTools for the web app, Jira for test cases and defects.

Entry criteria

- Q-01, Q-02, Q-04 and Q-05 answered and reflected in acceptance criteria.
- Build deployed to staging with the flag off by default.
- Mailer mock and seeded data available.

Exit criteria

- 100% of P1 cases passed; at least 90% of P2 passed.
- No open Critical or High defects.
- Rollback tested: switching the flag off stops new offers.

Eligibility decision table

Flag on	Plan Monthly/Quarterly	Check-ins < 4 (all members)	Expires ≤ 10 days	No balance, promo or corporate	Joined ≥ 45 days, no offer in 90 days	Plan type	Expected
Y	Y	Y	Y	Y	Y	Solo	**Offer 15%**
Y	Y	Y	Y	Y	Y	Duo	**Offer 25%**
Y	Y	Y	Y	Y	Y	Family	**Blocked by Q-01**
Y	Y	N	Y	Y	Y	Any	**No offer**
Y	Y	Y	N	Y	Y	Any	**No offer**
Y	Y	Y	Y	N	Y	Any	**No offer**
Y	Y	Y	Y	Y	N	Any	**No offer**
Y	N	–	–	–	–	Any	**No offer**
N	–	–	–	–	–	Any	**Not evaluated**

3. Test cases, prioritised

How I prioritise: impact × likelihood. **P1** covers money and trust (wrong member, wrong percentage, duplicates, failed renewal) and the main happy path; these run first and block the release. **P2** covers secondary rules, roles and channel consistency. **P3** covers performance at scale and less frequent paths.

ID	Scenario and data	Expected result	Technique	Priority
TC-01	**Eligible Solo member receives the offer end to end** 3 check-ins in 28 days, expires in 10 days, flag on.	Internal offer 15% created; POST to mailer with correct payload; 201 and offer_id stored; event logged; email received.	E2E + API	P1
TC-02	**Check-in boundary** Members with 3, 4 and 5 check-ins; class booked but not attended counts as 0.	Only the member with 3 check-ins is eligible.	Boundary values	P1
TC-03	**Expiry boundary** Memberships expiring in 9, 10, 11 days and expired yesterday.	9 and 10 eligible; 11 and expired are not.	Boundary values	P1
TC-04	**Flag off** Eligible member, winback_offer_enabled = false.	No evaluation, no offer, no call to the mailer.	Functional	P1
TC-05	**Duo plan, both inactive / one active** Two Duo memberships: both under 4 check-ins; one member with 6.	First gets 25%; second gets no offer.	Decision table	P1
TC-06	**No stacking with an active promotion** Eligible member with an active 10% promo renews.	No win-back offer created; checkout applies one discount only.	Business rule B7	P1
TC-07	**One offer every 90 days** Last offer sent 89 and 90 days ago.	89 days: no new offer. 90 days: new offer.	Boundary values	P1
TC-08	**Outstanding balance** Eligible member with an unpaid balance of $1.	No offer. Retest after payment once Q-09 is answered.	Business rule B1	P1
TC-09	**Mailer contract** Inspect the request sent to the mock.	member_id int, plan_type enum, discount_percent int, expires_on ISO date, source "winback"; 201 returns offer_id.	API contract	P1
TC-10	**Mailer fails or is slow** Mock returns 500, 429 and a 2.5 s timeout.	Error logged in /api/v2/audit/errors with member and offer IDs; no duplicate offer; retry follows Q-04.	Negative, API	P1
TC-11	**Daily job runs twice** Trigger the job twice on the same day.	One offer and one email per member.	Idempotency	P1
TC-12	**Accept the offer from the email** Tap the link on a phone with the app installed.	App opens the renewal with 15% applied; status "Renewed with discount" in both apps.	E2E mobile	P1
TC-13	**Regular renewal unchanged** Member without an offer renews.	Full price; no win-back data on the receipt.	Regression	P1
TC-14	**Offer accepted after expiry** Open the email link after the expiry date.	Discount not applied; clear message with the regular renewal option.	State transition	P2
TC-15	**Corporate membership and new member** Corporate member; members who joined 44 and 45 days ago.	Corporate and 44 days excluded; 45 days eligible.	Business rules B2, B6	P2

ID	Scenario and data	Expected result	Technique	Priority
TC-16	**Front desk renewal** Front desk renews a member with a pending offer.	Behaviour per Q-07; front desk cannot edit the percentage.	Role-based	P2
TC-17	**Admin cannot grant discounts** Admin calls POST /api/v2/offers directly.	403; no offer created; attempt logged.	Security	P2
TC-18	**Same offer data in both apps** Compare staff web app and member app.	Same percentage, expiry date and status.	Integration	P2
TC-19	**Email copy matches the plan** Duo member receives the email.	Email shows 25% (fails today, see Q-02).	Content	P2
TC-20	**Marketing opt-out** Eligible member who opted out of promotional email.	Behaviour per Q-08.	Compliance	P2
TC-21	**Mailer response time** Batch of 300 eligible members against the mock with production-like latency.	95% of calls under 2 s; no throttling errors lost.	Performance	P3
TC-22	**Link opened without the app** Tap the email link on a phone without the app.	Store page or web fallback; offer kept after install.	E2E mobile	P3

4. Three metrics to judge the release

Risk coverage

P1 and P2 cases passed over planned, with every business rule and acceptance criterion traced to at least one case.

Target: 100% P1, 90% P2, 100% rules traced.

Offer accuracy in production

Share of offers in the first two weeks sent to an eligible member, with the right percentage and no duplicate. Checked with a daily SQL audit.

Target: 99.5% or higher.

Defect leakage

Defects found after release over all defects found for the feature, during the first 30 days.

Target: 5% or less, no Critical.

> Business impact (renewal rate of inactive members) is tracked by Product. These three metrics answer a narrower question: did we ship what we meant to ship, safely?